

NATIONAL TEACHERS COLLEGE
School of Arts, Sciences, and Technology — BSIT Program

NTC_CC105 – DATA STRUCTURE W/LAB

WEEK 5 LABORATORY INSTRUCTION

Laboratory Activity 5

Instructor	Prof. Darwin C. Vargas, MIT
Program	BSIT
Academic Year	2026–2027
Programming Language	C++

1. Student Name	Quintela, Nicole M. / Serdenia Rihanna Shealya
2. Student Number	425001249 / 425002643
3. Section	2.4 BSIT
4. Date	Oct 5, 2026
5. Laboratory Activity No.	Laboratory Activity 5
6. Laboratory Title	Linear Architecture Synthesis & Peer Code Audit

PART F: Final Testing & Optimization Comparison Table

Test Case ID	Target Feature	Outcome Before Refactoring	Outcome After Refactoring	Engineering Performance Improvement / Observation
TC-01	Basic Element Insert	Function with Globals	Function with encapsulated class	Better organization; prevents accidental data changes
TC-02	Dynamic Memory Resize	Memory Leak	Old buffer Released	No resize Leak
TC-03	Traversal Output	Off-by-one read	Exact '0...n-1' traversal	No garbage Read
TC-06	Element Deletion	Redundant Nested loop	Single pass O(n) shift	Cleaner execution
TC-08	Upper Boundary Check	Possible corruption/crash	Throws 'out_of_range'	Safe rejection
TC-10	Program Cleanup Leaked	Heap leak	Destruction releases heap	AddressSanitizer confirms 0 bytes leaked on exit.